APPLICATION BRIEF · DESALINATION

Desalination pumping and energy recovery

Match the pump, membrane duty and complete plant energy boundary.

Reverse osmosis feed pumping, brine energy recovery and hydraulic actuation perform different jobs. A useful proposal identifies each duty, then sizes the selected equipment from the feed conditions and verified operating point.

Typical duties to assess

Duplex seawater feed pumping · RO Power Centre energy recovery · Modular treatment-train planning · Plant valve and gate actuation

Equipment and project scope

Select from water-service pumps, motors, cylinders, control valves and hydraulic power units. For RO, assess duplex seawater pumps and Power Centre configurations. The quotation defines equipment, interfaces, documentation and integration responsibilities.

What to establish

Send net production, operating hours, feed salinity and temperature, recovery ratio, membrane duty and inlet conditions. Include pretreatment, flushing, concentrate handling, storage, auxiliaries and shared dependencies in the plant scope.

Evidence and boundaries

For illustration: 20 m³/day (5,283 US gal/day) over 20 hours requires 1 m³/h (4.40 US gpm) of permeate. At an assumed 40% membrane recovery, membrane feed is 2.5 m³/h (11.0 US gpm), before other losses. This is a water-balance example, not a package rating.

Start with the duty

Send the asset, operating problem, location, required timing and available drawings. We will assess the equipment scope and the evidence needed for a quotation.

Proposed applications are not customer installations or approvals. Confirm ratings and documentation for the selected equipment and project.